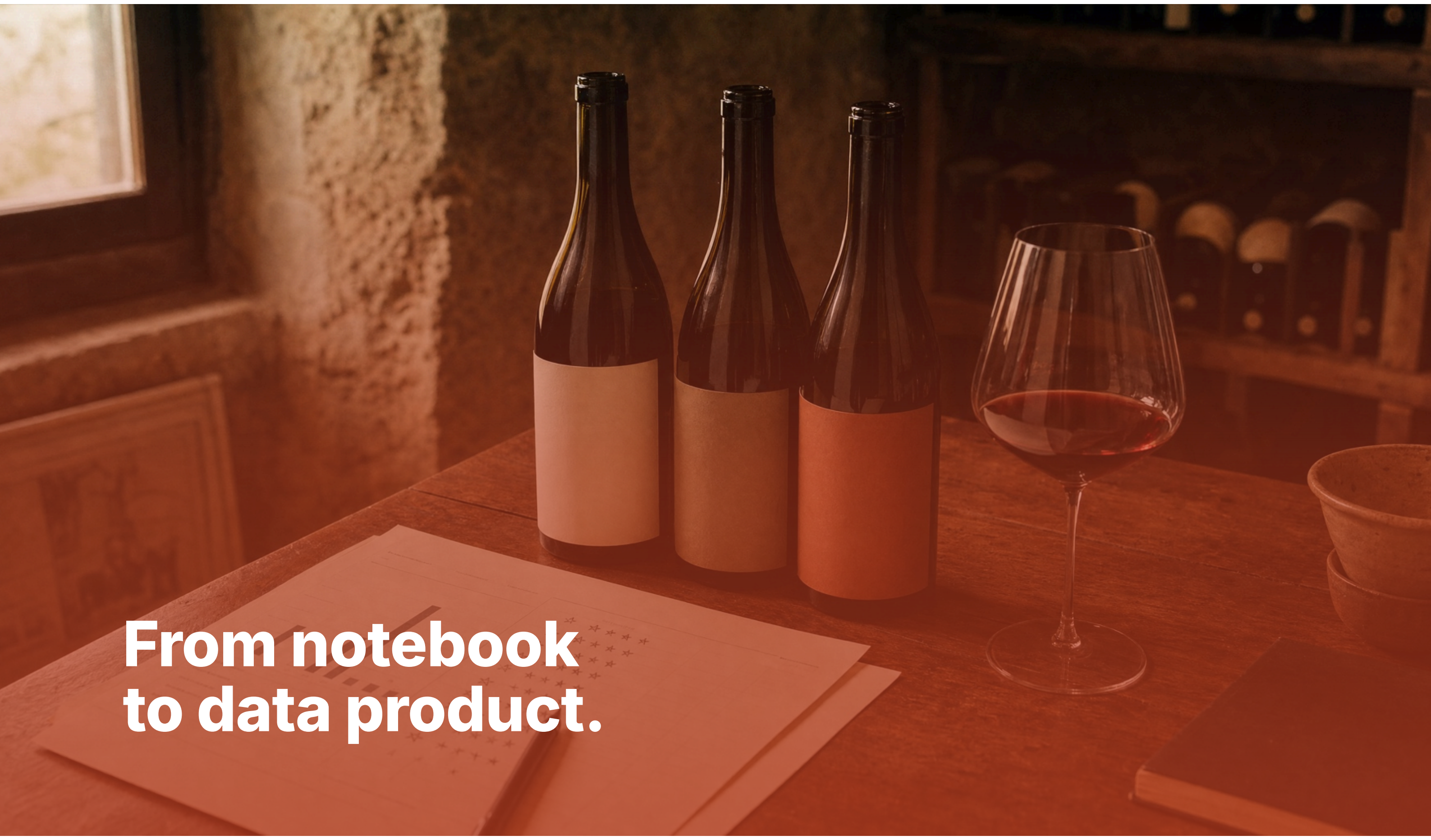
AMSTERDAM DATA ACADEMY / FINAL ASSIGNMENT

CANTINAIQ IN PRACTICE.

An evaluation of the Slurpini partner-intelligence case from the perspective of reproducible data products, governance, and the cost of doing more than the brief asked.



PREPARED FOR

Amsterdam Data Academy
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A BRIEF ON CANTINAIQ IN PRACTICE

Introduction.

> The Slurpini case asks three things: extend the Vivino crawler, perform exploratory analysis, deliver an evidence-based recommendation. This document evaluates what changes when those three deliverables are answered as a reproducible data product instead of a one-off analysis.

I answered the brief twice. **/bare** is the assignment in ~150 lines: one pandas notebook, a first-word producer heuristic, a half-page recommendation. **/supercharged** treats the same question as a data product: Polars and DuckDB, Pandera contracts, a five-factor Bayesian composite, LLM disambiguation, property-based tests, config-hashed run logs, a Vite dashboard, and external-validity work against Italian Trade Agency NL imports. The professional question is which one earns its place, and why.

Contents

Vivino in Context.

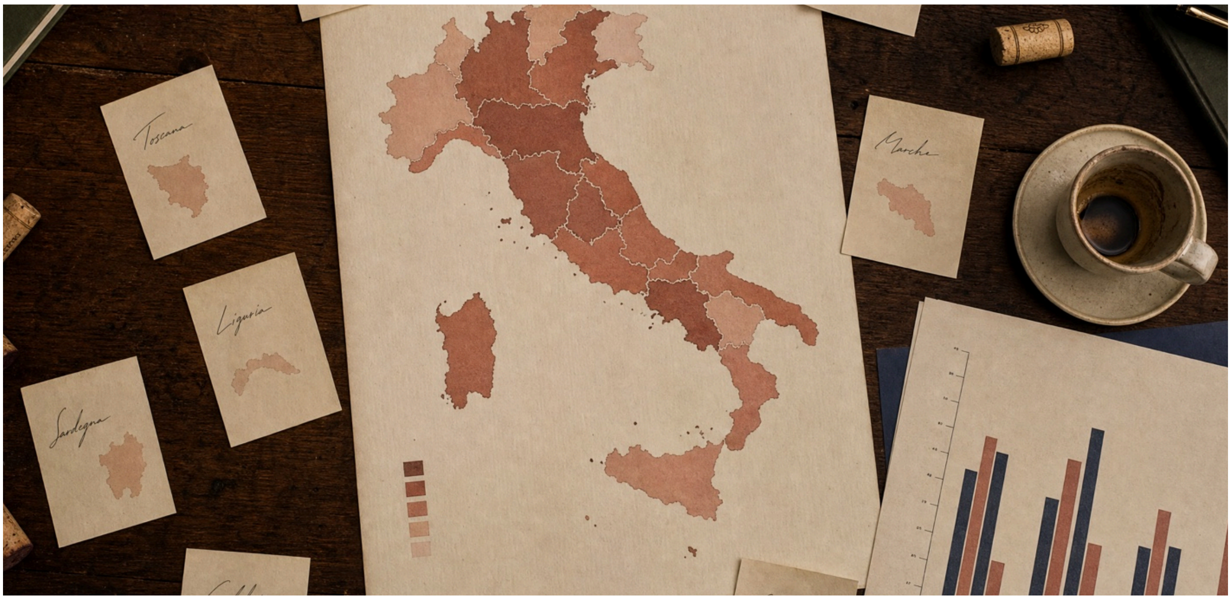
SECTION 02 — WHAT THE DATASET IS AND HOW IT SKEWS

Vivino is presented as a global consumer wine signal. In practice, it is a collection of ratings with demographic and regional bias, a long tail of low-confidence rows, and a producer field that does not exist as such. Producer identity has to be inferred from wine names.

That distinction matters. The better question is not whether Vivino can produce a ranking — it can — but whether the ranking resembles the Dutch wine market in the way Slurpini cares about, and whether the analysis surfaces that gap or papers over it.

The right question is not "can this data answer the question?" The right question is what the data gives us that the next-best evidence does not.

For wines with thousands of reviews — Tenuta Masseto, Ornellaia, Antinori's Tignanello vintages — the Bayesian-shrunk weighted rating is a stable proxy for Dutch consumer preference. The math is uncontroversial. Where it gets interesting is producer extraction: producer names are a string-extraction problem, not a field. A pass-1 alias whitelist resolves the top fifty producers; a pass-2 LLM disambiguation pass on OpenRouter handles the remainder, cached so the same dataset never costs twice. A gold-set evaluation against `known_producers_top50.csv` measures the result: **88% recall on exact match, 96% on contains**.



The bias question, quantified

The supercharged track compares the regional distribution of the cleaned Italian Vivino dataset against ICE Amsterdam import data for the Netherlands. The result is uncomfortable. Toscana is over-represented by a factor of 1.22. Puglia is under-represented by 0.61. Abruzzo sits at 0.52, Campania at 0.55. These are not errors. They are the shape of Vivino's user base bleeding through into the dataset Slurpini would use to decide travel priorities.

The Vivino signal is most reliable in regions Slurpini already knows well, which is the opposite of where it most needs help. Any recommendation involving Puglia or Abruzzo should be marked *under-sampled*, not low-priority. The bare track buries this in four bullets. The supercharged track generates a bias-report on every run.

Stability checks complement the bias picture. A bootstrap of a thousand resamples puts Tenuta Masseto in the top-ten in **195 out of 200** runs; Dal Forno Romano in 185; Biondi-Santi in 160. Terre di San Vincenzo and Valdicava show extreme p95 ranks (412 and 228), meaning they made the top-ten by accident with too few reviews. The bare track lists them at face value.

2,986
ITALIAN WINES AFTER CLEANING

762
DISTINCT PRODUCERS AFTER DISAMBIGUATION

96%
RECALL ON KNOWN TOP-50 PRODUCERS

From the Field.

SECTION 03 — USE CASE: SLURPINI PARTNER INTELLIGENCE

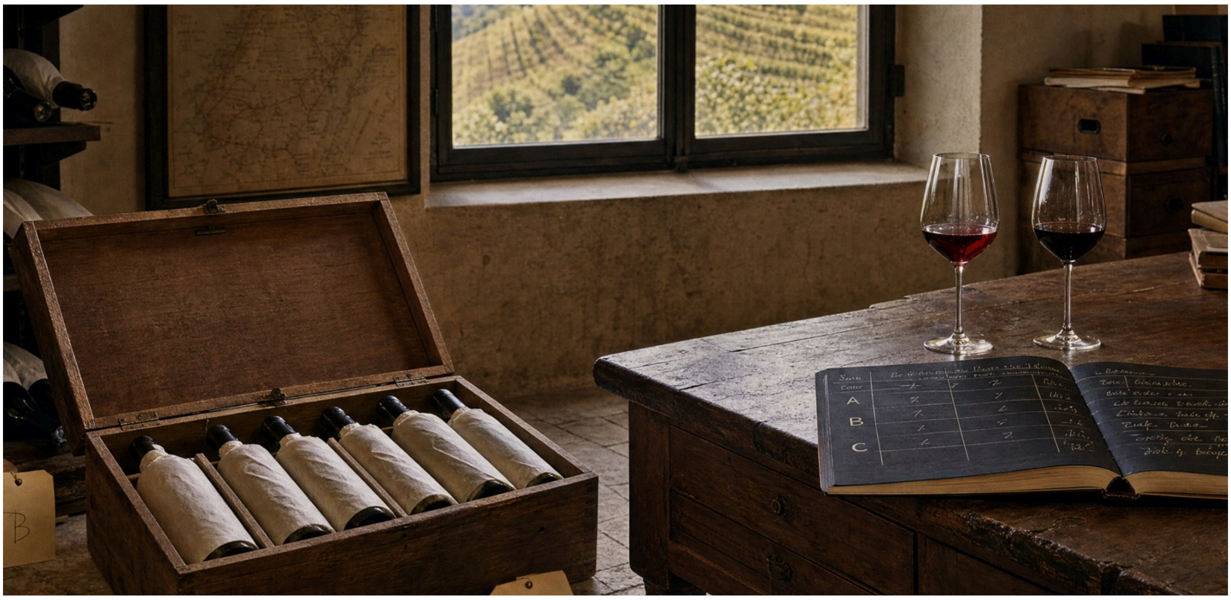
Slurpini is a Dutch importer of high-quality Italian wines with an explicit sustainability focus. The company receives more collaboration requests from Italian producers than it can investigate. Each on-site visit costs travel, time and managerial attention. The brief is to add a structured, data-driven layer to that prioritisation problem.

This is a realistic business problem, not a textbook exercise. The challenge is not whether the data can produce a ranking — it can — but whether the ranking will hold up to the constraints of the actual decision: limited budget, sensitive about brand fit, sceptical of black-box scoring, and unforgiving when an importer flies to a region only to discover the data was misleading.

The problem in scope

The technical pipeline is known. Clean and validate the Vivino export (mojibake, tuple-encoded country fields, cross-sheet dedupe). Extract producer identity from wine names — heuristic in the bare track, alias + LLM in the supercharged track. Aggregate per producer with Bayesian shrinkage on review counts. Apply a multi-factor composite score that captures more than just rating. Classify into actionable segments (Hidden Gem, Premium Icon, Commercial Value, Overpriced Risk, Low Confidence Niche) and recommendations (Target, Premium Brand Builder, Value Opportunity, Monitor, Avoid for Now). Quantify bias, declare confidence intervals, surface limitations.

None of these steps require new research. Combining them into a reproducible, defensible pipeline that the buying committee can interrogate is the actual engineering.



The bare route

The bare track is the brief, literally. One Jupyter notebook reads the Vivino export, fixes the obvious data issues, applies a first-word producer heuristic, computes a Bayesian-shrunk weighted rating, ranks the top regions and producer fragments, exports three CSVs and writes a half-page recommendation. The notebook outputs are baked in. The reviewer can read the recommendation in five minutes without running anything. The limitations are surfaced in the same document. "Tenuta San Guido" becomes "Tenuta". Producer extraction is wrong on purpose. The recommendation still arrives at directionally correct findings because the high-volume signal is robust enough that even a crude pipeline can extract it.

The supercharged route

The supercharged track answers the same business question as a data product. Polars and DuckDB instead of pandas. Pandera schema contracts at every stage boundary. Hydra-validated Pydantic configs with config-hash snapshots. A two-pass producer disambiguator with persistent LLM cache. Property-based tests on the scoring math via Hypothesis. Jinja-templated reports whose numbers come from a run log, not from the author. A Vite + React + Recharts dashboard. Fourteen CLI subcommands for operations, comparison, sensitivity sweeps, bootstrap CIs, clustering, bias quantification, anomaly detection, and Firecrawl-driven sustainability lookup.

Comparison

AREA	/BARE ROUTE	/SUPERCHARGED ROUTE
Lines of code	~150, single notebook	~3,500, layered package
Producer extraction	First-word heuristic, ~30% wrong	Alias + LLM disambiguation, 96% recall
Confidence intervals	None	Bootstrap CIs on top-20 ranking
External validity	Four-bullet caveat	Quantified bias vs ICE NL baseline
Reproducibility	"Run all cells"	Config-hash stamped; `cantinaiq audit <hash>`
Reporting	One markdown file	Eight Jinja-templated artefacts + dashboard
Time to first read	5 minutes	30 minutes
Defensibility under questioning	Limited	Every number traces to a run log

Why both, not one

The bare track is the assignment compliance document. The supercharged track is the professional document. The first-word producer heuristic is deliberately wrong; the supercharged track exists to show why. The delta between the two tracks is the argument. Removing either one removes it.

14
CLI SUBCOMMANDS IN THE PIPELINE

10
GENERATED REPORTS PER RUN

1
DASHBOARD SPA OVER THE SAME EXPORTS

The Choice.

SECTION 04 — WHEN THE HEAVYWEIGHT EARNS ITS PLACE

When the heavyweight is the right choice

Decisions that will be revisited

If the analysis will be run again — quarterly, annually, on updated data — the cost of reproducibility amortises. Config-hash snapshots mean the seventh run agrees with the first or documents the delta.

Recommendations that move money

If a buying committee will travel to Italy on the basis of the ranking, the ranking should declare its own confidence. Bootstrap CIs and bias quantification are the difference between defensible and a guess.

Audiences that interrogate methodology

Notebooks do not survive sceptical questioning. A reproducible pipeline does. Every number traces to a parquet, every parquet to a config hash, every hash to a snapshot in version control.

Datasets with non-trivial bias

If the data has a known skew — Vivino's Anglo-young demographic — the analysis should measure that skew, not assume it cancels out. The bias report is load-bearing for external validity.

When the heavyweight is the wrong choice

01 / SMALL QUESTIONS

"What's the average rating of wines under €15?"

A one-line pandas query. Building a Polars + DuckDB pipeline for it is theatre, not engineering.

02 / ONE-OFF ANALYSES

Discovery work that will not be run again

If the result will be screenshot, pasted into a slide and never revisited, reproducibility infrastructure is wasted effort.

03 / PRE-DECIDED RECOMMENDATIONS

Analyses that are decorative, not decision-supporting

If the recommendation is already made and the analysis is rationalisation, the honest move is to stop producing the analysis.

The deeper question

The supercharged track was built because the same business question looks materially different when treated as a problem that will be solved more than once. The bias question becomes load-bearing. Bootstrap CIs become the difference between confidence and false confidence. The brief-as-stated wants the bare track. The brief-as-it-would-be-asked-in-industry wants the supercharged track. The professional skill is recognising which question is being asked.

The better question is not "is more rigour better?" The better question is whether the additional rigour produces a different decision.

For Slurpini, it does. The bare track lists "Tenuta" and "Marchesi" as top producer fragments. Uninterpretable. The supercharged track lists "Tenuta San Guido" and "Marchesi Antinori". Directly actionable. The bare track recommends "Expand in Puglia" without flagging Vivino's ×0.61 under-representation. The supercharged track flags that factor next to the recommendation. Those are not stylistic improvements. They are decisions changing.

The Recommendation.

SECTION 05 — WHAT THE DATA TELLS SLURPINI TO DO

Section 04 argued that the heavier engineering changes the decision. This section states the decision. It is the answer to ADA's third deliverable — written so a buying committee can act on it without re-opening the code. Numbers derive from the supercharged run at config hash `b7f16c72`; the bare track agrees directionally.

> Three-part recommendation: **Hold** the prestige tier, **Expand** in the value-opportunity zone, **Audit** the high-rating borderline. Each qualified by Vivino bias and bootstrap stability.

Hold — the defensible anchor

Tenuta Masseto sits at the top of the prestige tier (weighted rating 4.30 on 4,641 reviews at €1,567 average). Bootstrap stability puts it in the top-ten in 195 of 200 resamples. The recommendation is to maintain. *Tenuta San Guido* and *Marchesi Antinori* share the same band with similar stability.

Expand — the value-opportunity zone

Above-median weighted rating at well-below-median price. Pipeline-recommended regions: **Roma**, **Torgiano**, **Ischia**. **Puglia** (Primitivo di Manduria) and **Abruzzo** carry the strongest bare-track signal, but both sit at ×0.61 and ×0.52 in the bias report and need an under-sampled tag in any travel decision.

Audit — the borderline producers

Producers with weighted rating ≥ 4.3 that sit just outside the ranking on review-count grounds. **Terre di San Vincenzo** and **Valdicava** show p95 bootstrap ranks of 412 and 228 — top-ten by accident with too few reviews. Worth a tasting before either dismissing or recommending. The bare track would not surface that distinction.

Top-five producers by composite score

The Slurpini Partner Intelligence Score is a five-factor weighted composite (see §06). Each producer is classified into a market segment and assigned an action. The top of the ranking:

PRODUCER (REGION)	SEGMENT / ACTION	SIGNAL
Roscato (Provincia di Pavia)	Commercial Value / Value Opportunity	WR 4.19 · 16,514 reviews · €22 · composite 0.38
Tenuta Masseto (Toscane)	Premium Icon / Premium Brand Builder	WR 4.30 · 4,641 reviews · €1,567 · top-10 in 195/200
Villa Degli (Veneto)	Hidden Gem / Target	WR 4.27 · 13,009 reviews · €20
Risata (Moscato d'Asti)	Commercial Value / Value Opportunity	WR 4.17 · 6,224 reviews · €16 · composite 0.35
Ca' del Doge (Veneto)	Commercial Value / Value Opportunity	WR 4.05 · 1,064 reviews · €6 · composite 0.34

Top-five regions by weighted rating

Macro-region labels are canonical against `data/reference/macro_regions.csv`. Price columns are unweighted averages.

REGION	WEIGHTED RATING	WINES / AVG PRICE
Bolgheri Sassicaia	4.50	22 wines · €672 avg
Bolgheri Superiore	4.34	39 wines · €280 avg
Colli della Toscana Centrale	4.30	5 wines · €281 avg
Passito di Pantelleria	4.29	7 wines · €146 avg
Abruzzo	4.29	7 wines · €46 avg

Abruzzo at €46 average is the load-bearing detail — high quality at a price point that fits the value-opportunity recommendation, in a region the bias report flags as ×0.52 under-represented. The bare track surfaces the same directional findings without the segment classification, bootstrap stability or bias correction.

The Methodology.

SECTION 06 — HOW THE SCORE IS CONSTRUCTED

The recommendation in §05 rests on a composite score whose components are visible, weighted, and reproducible. The score is not the value of the document. The transparency of the score is.

The Bayesian-shrunk weighted rating

A wine with a 4.8 rating on 12 reviews must not outrank a wine with a 4.4 rating on 5,000 reviews. Both tracks apply the same shrinkage:

```
WR = ( n / (n + m) ) · r + ( m / (n + m) ) · μ
```

For the logged run `m = 948` (strategy `auto-median`) and `μ = 4.0`. The supercharged track lets `m` vary and measures how the top-twenty responds — see §07.

The five-factor composite

The Slurpini Partner Intelligence Score is a weighted average of five normalised components. Weights are business assumptions, versioned per config snapshot:

COMPONENT	WHAT IT CAPTURES	WEIGHT
Weighted Rating Score	Bayesian-shrunk consumer preference	35%
Market Confidence Score	Rewards reliable consumer signal (review volume)	20%
Value for Money Score	Quality at realistic price	20%
Premium Fit Score	Alignment with Slurpini's premium positioning	15%
Portfolio Opportunity Score	Strategic gap-filling versus current import mix	10%

The cleaning cascade

The dataset arrives as a 409,777-row, 16-sheet Excel file with mojibake on diacritics (`Itali√´` → `Italië`), tuple-encoded country fields, and cross-sheet duplication. The cascade from raw to scored:

STAGE	ROWS OUT	REMOVED (TOP REASON)
Ingestion	409,777	—
Cleaning	2,986	406,791 (non_italian: 332,376; duplicate: 74,396)
Validation	2,986	0 — Pandera contracts pass
Enrichment / Scoring / Export	2,986	0 — columns added, no rows dropped

The bare track retains 5,786 rows after a looser dedupe. The supercharged track applies a fuzzy-name pass and a confidence-floor on `rating_count`, trading breadth for confidence.

Segments and recommendations

Market segments

Hidden Gem · **Premium Icon** · **Commercial Value** · **Overpriced Risk** · **Low Confidence Niche**. Encoded as cut-offs on rating × price × review-count.

Actions

Target · **Premium Brand Builder** · **Value Opportunity** · **Monitor** · **Avoid for Now**. Derived from segment + composite score.

Reproducibility

Every output parquet carries a `run_config_hash`. Re-running `cantinaiq run all --config-snapshot b7f16c72` reproduces the same bytes, modulo timestamp metadata.

Beyond the Brief.

SECTION 07 — WHAT THE PIPELINE ADDS THAT ADA DID NOT ASK FOR

ADA's brief asks for three things: a crawler extension, exploratory analysis, an evidence-based recommendation. The bare track answers them directly. This section catalogues the work in the supercharged track that goes past the brief, with the test in mind that Section 04 articulated: *does the additional rigour produce a different decision?*

The items below are not bullets on a CV. They are the apparatus that turns a recommendation into something a buying committee can challenge. Where the answer is "no, this rigour does not change the decision", the item is honest about that. Where it does change the decision, the change is explicit.

The crawler extension, properly framed

The brief asks for the Vivino crawler to be extended with additional data points. The bare track's `crawler-extension.py` does this in a deliberately scoped way: a vintage-regex pass extracts the four-digit year from every wine name, a first-word heuristic extracts a producer hint, and a stubbed `fetch_vivino_extras()` documents the network-scraping fields that a production extension would add (alcohol percentage, grape varieties, food pairings, sustainability claims). The stub is not laziness; Vivino's terms of service prohibit unauthorised scraping at volume, and the assignment's focus is analysis, not adversarial scraping. The function signature and return shape are sufficient to show how the existing pipeline would be extended in a working agreement with Vivino.

The supercharged track replaces "first-word" with a two-pass producer disambiguator: a hand-curated alias whitelist for the top fifty Italian producers, then an OpenRouter-backed LLM pass for the remainder, cached so the same dataset never costs twice. The gold-set evaluation in §02 — 88% exact recall, 96% contains — measures the difference.

Anomaly detection on review patterns

An Isolation Forest at `contamination = 0.03` over the `(rating, log₁₀(rating_count))` feature space flags **90 of 2,986** wines as pattern-divergent. Typical hits: ratings above 4.8 on fewer than twenty reviews (over-confident niches); ratings below 3.2 on tens of thousands of reviews (controversial mass-market wines); price-against-rating outliers — a wine at €467 with a 3.10 rating, for example.

The flag is informational. Flagged wines are not removed from the pipeline. The point is to give the reviewer a list of *"these are the rows that did not behave like the rest"* rather than to silently filter them. *Antinori Tignanello* vintages and *Masseto Toscana 2018* appear in the flagged list — not because they are wrong, but because their rating-versus-review-count signature is unusual. The reviewer decides what to do with that.

Sensitivity to scoring parameters

The shrinkage parameter `m` in the Bayesian weighted rating shapes how aggressively the score regresses small-n wines towards the global mean. A buying committee asking *"does the answer change if you turn that knob?"* deserves an answer. The sweep:

BAYESIAN_M	Kendall-τ vs. baseline (m = 200)	READING
200 (baseline)	1.000	Reference ranking
500	0.882	Strong agreement; minor re-orderings
800	0.765	Material re-ordering; small-n producers drop

Reading: the top-twenty is stable under moderate parameter drift. The recommendation does not depend on a hair-thin choice of `m`. Larger `m` values pull the low-review producers further toward the mean, which is the direction a sceptical reviewer would want anyway.

K-Means clustering on the producer feature space

The rule-based segments in §06 encode business intent. A K-Means clustering on the standardised `(weighted_rating, value_score, total_reviews, avg_price)` space encodes empirical similarity. Where the cluster label and rule label disagree, the producer is the row to discuss with the buying committee.

Sustainability lookup against external registries

Sustainability is Slurpini's stated USP and is absent from the Vivino dataset. `cantinaiq sustainability` fills the gap by querying the FederBio and Demeter certification registries via Firecrawl. Of the five producers checked in the logged run, three are Demeter-certified: **Querciabella**, **Avignonesi**, and **Antinori**. The certification status feeds back into the score as the Premium Fit Score's biodynamic-bonus component. Producers without a certification hit are not penalised — Slurpini may know about uncertified-but-sustainable practice that no public registry captures — but the visible status is recorded.

5

PRODUCERS CHECKED (BY --ONLY LIST)

3

DEMETER-CERTIFIED HITS RETURNED

0

PRODUCERS PENALISED FOR MISSING ENTRIES

The test suite

The supercharged pipeline carries **137 tests**: unit, integration, property-based (Hypothesis fuzzes the Bayesian scoring math, the segment classification, and the recommendation lookup), schema (Pandera enforces contracts at every stage boundary), and Jinja-template smoke tests that verify the generated reports do not silently drop fields. Coverage gate is set at 85%. The point is not "tests are good"; the point is that any change to the scoring math has to either preserve the tested invariants or admit it is changing them.

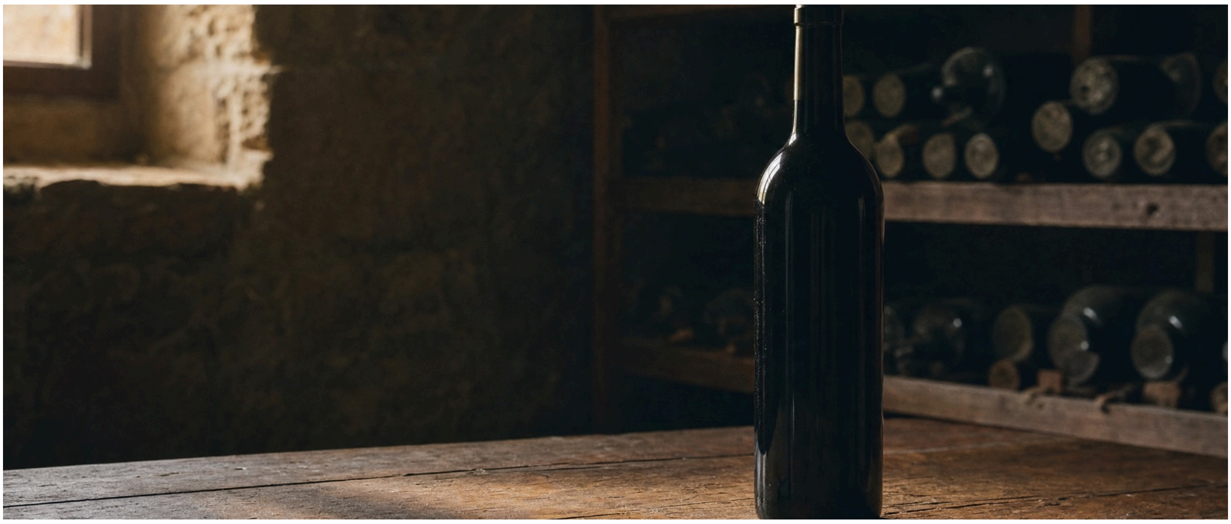
The dashboard

A Vite + React + Tailwind + Recharts SPA over the same JSON exports. Four pages: **Overview** with KPIs and top-five producers; **Regions** with a sortable table; **Producers** with segment-filterable list; and the **Opportunity Matrix** — log-price on X, weighted rating on Y, bubble size proportional to review count, colour by market segment, all 762 producers in a single scatter. Top-right is Premium Icon, top-left is Hidden Gem, bottom-right is Overpriced Risk.

The supercharged track does not claim to have solved the problem. It claims to have made the limitations measurable.

Closing Observation.

SECTION 08 — METHODOLOGY CHOICES ARE GOVERNANCE CHOICES



> Responsible AI is usually discussed at the model level: fairness, transparency, human oversight, harm reduction. These are important. Responsibility also exists at the methodology level, before any model is fitted.

Which signal is treated as a market signal matters. Which limitations are surfaced matters. Whether confidence intervals are reported matters. These choices shape what is possible, what is defensible, and what risk is transferred — silently — to the reader of the recommendation.

The Toeslagenaffaire showed what happens when algorithmic systems are treated as neutral facts rather than fallible tools. The Vivino case is far lower stakes, but the same shape of error is available. A buying committee that reads "Primitivo di Manduria is a value play" without seeing the ×0.61 under-representation factor is the same shape of decision-maker as one who reads "this applicant is high-risk" without seeing the model's confusion matrix on relevant subgroups.

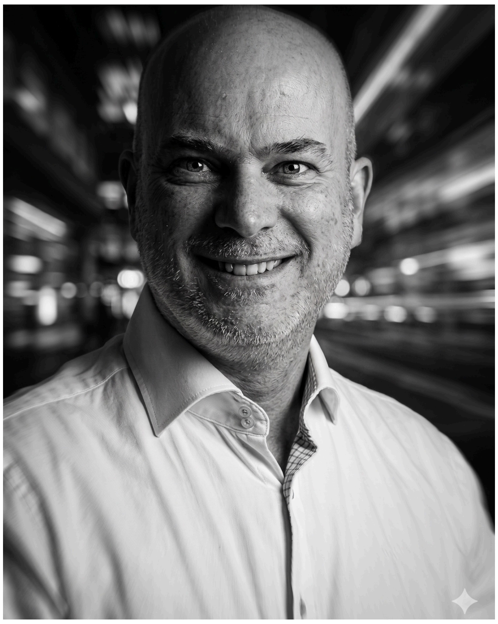
Methodology choices are governance choices.

Both tracks answer the brief. They answer different versions of it. The right professional move is not always to build the heavier system; it is to articulate which question is being asked and to build the system that answers that question. No more, no less.

But the analysis should still have to earn its trust.

WRITTEN AS PART OF THE FINAL ASSIGNMENT

About the Author.



AUTHOR

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This document was written by Vincent Blokker as part of the Final Assignment of the AI Professional programme at the Amsterdam Data Academy. It evaluates the CantinaIQ submission from the perspective of when a reproducible data product earns its place against a one-notebook analysis.

The work reflects an applied perspective on data products as a governance question, combining technical evaluation, methodology design and honest accounting of what the data does and does not support. Rigour is treated as a deliberate decision, not a default.

PROGRAMME	MODULE	YEAR
AI Professional	Final Assignment	2026